

Introduction To CS And Programming Using Python

Syllabus

Instructor Insights

Calendar

Readings

Materials by Lecture

Materials by Type

Calendar

In-person lectures were mandatory and occurred on Mondays and Wednesdays. Recitations were held on Fridays and were optional. Finger Ex. stands for “Finger Exercises” and “PS” stands for “Problem Set.”

WEEK #	MONDAY LECTURE	WEDNESDAY LECTURE	RECITATIONS
1	No lecture	Lecture 1 Introduction to Python: knowledge, machines, objects, types, variables, bindings, IDEs Readings: Ch 1, Ch 2.1–2.2 Lecture 1 Finger Ex. handed out PS 0 out (not graded)	Recitation 1
2	Lecture 2 Core Elements of Programs: strings, input/output, f-strings, operators, branching, indentation Readings: Ch 2.3–2.4 Lecture 2 Finger Ex. handed out Lecture 1 Finger Ex. due PS 1 out	Lecture 3 Program Flow: control flow, loops Readings: Ch 2.5–2.8 Microquiz 1 (30 minutes in-class on your computer) Lecture 3 Finger Ex. handed out Lecture 2 Finger Ex. due	Recitation 2
3	Lecture 4 Iteration, Simple Programs: guess and check, binary, fractions Readings: Ch 3.1, 3.3 Lecture 4 Finger Ex. handed out Lecture 3 Finger Ex. due	Lecture 5 Simple Algorithms: approximation method, floats Readings: Ch 3.2–3.3 Microquiz 2 (30 minutes in-class on your computer) Lecture 5 Finger Ex. handed out Lecture 4 Finger Ex. due PS 1 halfway hand-in due	No recitation
4	Lecture 6 Simple Algorithms: bisection search, Newton-Raphson Readings: Ch 3.4–.5 Lecture 6 Finger Ex. handed out Lecture 5 Finger Ex. due	Lecture 7 Functions: decomposition, abstraction, specifications Readings: Ch 4.1–4.2 Microquiz 3 (30 minutes in-class on your computer) Lecture 7 Finger Ex. handed out Lecture 6 Finger Ex. due	Recitation 3
5	Lecture 8 Functions: environments, scope, functions as objects Readings: Ch 4.3–4.6 Lecture 8 Finger Ex. handed out Lecture 7 Finger Ex. due	Lecture 9 Tuples and Lists Readings: Ch 5.1–5.3 Microquiz 4 (30 minutes in-class on your computer) Lecture 9 Finger Ex. handed out Lecture 8 Finger Ex. due PS 2 out PS 1 due	Recitation 4
6	No lecture	Lecture 10 List Operations, Mutability: mutation, aliasing, tricky examples with loops over L Readings: Ch 5.3–5.5 Lecture 10 Finger Ex. handed out Lecture 9 Finger Ex. due PS 2 halfway hand-in due	Recitation 5
7	Lecture 11 Aliasing and Cloning, List Comprehensions Readings: Ch 5.3–5.5 Microquiz 5 (30 minutes in-class on your computer) Lecture 11 Finger Ex. handed out Lecture 10 Finger Ex. due	Lecture 12 More Functions as Objects, Keyword Arguments, Default Arguments, Debugging: glass box/black box testing, examples Readings: Ch 4.4, Ch 8 Lecture 12 Finger Ex. handed out Lecture 11 Finger Ex. due PS 3 out PS 2 due	Recitation 6
8	Lecture 13 Exceptions, Assertions Readings: Ch 9 Microquiz 6 (30 minutes in-class on your computer) Lecture 13 Finger Ex. handed out Lecture 12 Finger Ex. due	Lecture 14 Dictionaries: keys, values, mutability, iteration over a dict, examples Readings: Ch 5.7 Lecture 14 Finger Ex. handed out Lecture 13 Finger Ex. due PS 3 halfway hand-in due	Recitation 7
9	Lecture 15 Recursion: iteration vs recursion, inductive reasoning Readings: Ch 6.1 Microquiz 7 (30 minutes in-class on your computer) Lecture 15 Finger Ex. handed out Lecture 14 Finger Ex. due	Lecture 16 Recursion: Fibonacci, Fibonacci with a dict, recursion on non-numerics, recursion on lists, Towers of Hanoi (extra) Readings: Ch 6.2–6.4 Lecture 16 Finger Ex. handed out Lecture 15 Finger Ex. due PS 4 out PS 3 due	Recitation 8
10	Lecture 17 Object Oriented Programming: data abstraction, class def, class instances, methods Readings: Ch 10.1 Microquiz 8 (30 minutes in-class on your computer) Lecture 17 Finger Ex. handed out Lecture 16 Finger Ex. due	Lecture 18 Object Oriented Programming: dunder methods, examples Readings: Ch 10.1 Lecture 18 Finger Ex. handed out Lecture 17 Finger Ex. due PS 4 halfway hand-in due	No recitation
11	Lecture 19 Inheritance: hierarchies, subclasses, using inherited methods, examples Readings: Ch 10.2 Lecture 19 Finger Ex. handed out Lecture 18 Finger Ex. due	Lecture 20 Inheritance: more examples Readings: Ch 10.4 Lecture 20 Finger Ex. handed out Lec19 Finger Ex. due PS 5 out PS 4 due	Recitation 9
12	Lecture 21 Complexity: measuring efficiency, timing programs, counting operations Readings: Ch 11 Microquiz 9 (45 minutes class on your computer) Lecture 20 Finger Ex. due No Lecture 21 Finger Ex.	No lecture	No recitation
13	Lecture 22 Complexity: Big-Oh notation, Big-Theta notation, complexity relations and classes, calc-complexity Readings: Ch 11 Lecture 22 Finger Ex. handed out	Lecture 23 String and List Examples, Analyzing Complexity, Search: indirection, linear search, bisection search Readings: Ch 12.1 Lecture 23 Finger Ex. handed out Lecture 22 Finger Ex. due PS 5 halfway hand-in due	Recitation 10
14	Lecture 24 Sort: bogo, bubble, selection, merge sort Readings: Ch 12.2 Microquiz 10 (30 minutes in-class on your computer) Lecture 23 Finger Ex. due No Lecture 24 Finger Ex.	Lecture 25 Visualization Library Readings: Ch 13 PS 5 due No Lecture 25 Finger Ex.	No recitation
15	Lecture 26 Extras: lists in memory, hashing, simulations, wrap-up Readings: Ch 12.3, Ch 17 No Lecture 26 Finger Ex.	No lecture	No recitation

Course Info

INSTRUCTOR

[Dr. Ana Bell](#)

DEPARTMENTS

[Electrical Engineering and Computer Science](#)

AS TAUGHT IN

Fall 2022

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